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Heterogeneous OS Benchmarking for Real-Time Edge Control on Asymmetric Hardware

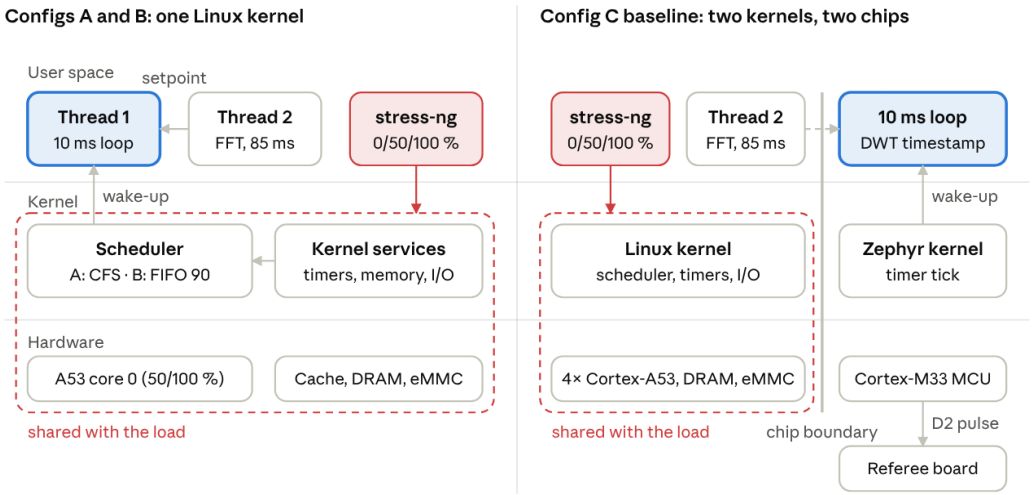
1. Project summary

We test whether a 10 ms control loop keeps its deadlines under heavy Linux load in three OS setups on one Arduino UNO Q. The board pairs a Qualcomm QRB2210 (4× Cortex-A53, Debian 13 Linux) with an STM32U585 (Cortex-M33 at 160 MHz, Zephyr RTOS). The test workload is a Prescriptive Maintenance pipeline: CWRU bearing vibration frames → 1024-point FFT → motor speed setpoint.

2. Experiment setup

On Linux, a 10 ms control thread wakes on an absolute schedule (clock_nanosleep, TIMER_ABSTIME) and logs each wake-up time to RAM. A second thread runs a 1024-point FFT on replayed CWRU bearing frames every 85.33 ms and updates a shared setpoint atomically.

- Config A: default Linux scheduler (SCHED_OTHER).
- Config B: SCHED_FIFO priority 90 with mlockall.
- Config C baseline: the same 10 ms loop on the MCU, timestamped with the DWT cycle counter and cross-checked by a second UNO Q timing its output pulses. So far it is measured without IPC (baseline); the Linux → MCU setpoint link is on the next experiment.



Each run lasts 10,000 cycles (100 s) at 0, 50 or 100 % stress-ng load (CPU, cache, disk and memory workers, all pinned to core 0). At 50/100 %, the Linux threads share that core. The CPU governor is fixed at 2.016 GHz.

Cycle k is scheduled at $R_k = t0 + k \cdot 10$ ms (t0: run start), and its lateness is $L_k = t_k - R_k$ (t_k: measured wake-up time). We report σ_L, p99.9, worst-case lateness (WCL) and the deadline miss ratio (DMR: share of cycles with $L_k > 1.0$ ms or > 0.2 ms). For Config C, lateness is measured from the best cycle after fitting the period, so compare only σ_L, p99.9 and WCL across configs.

3. Results and explanation

Worst-case lateness reaches 6.12 ms for CFS (at 50 % stress; 5.80 ms at 100 %), stays under 0.87 ms for SCHED_FIFO, and stays under 50 µs for the MCU loop. All 9 runs completed 10,000 cycles (100 s) without a crash.

Config	Stress	WCL (µs)	Jitter σ_L (µs)	p99.9 (µs)	DMR 1.0 ms	DMR 0.2 ms
A - CFS	0 %	634.2	30.4	400.9	0.00 %	96.93 %
A - CFS	50 %	6,119.7	482.1	4,178.5	2.76 %	6.83 %
A - CFS	100 %	5,804.9	1,041.0	4,699.5	13.91 %	18.46 %
B - FIFO 90	0 %	478.2	34.0	316.0	0.00 %	1.94 %
B - FIFO 90	50 %	865.2	38.4	412.7	0.00 %	1.55 %
B - FIFO 90	100 %	759.2	39.0	366.0	0.00 %	1.73 %
C - MCU baseline	0 %	10.3	2.6	10.3	0.00 %	0.00 %
C - MCU baseline	50 %	44.6	12.2	44.4	0.00 %	0.00 %
C - MCU baseline	100 %	49.1	13.4	49.0	0.00 %	0.00 %

Linux is slower at idle than under load: median lateness is 224 µs (A) and 171 µs (B) at 0 % stress, but 44–51 µs for B at 50 and 100 %. Because A's idle median is already above 200 µs, 97 % of its idle cycles miss the 0.2 ms tier. The likely cause is wake-up from deep CPU idle states when nothing keeps the core busy (the performance governor fixes the clock frequency, not the idle states); we have not confirmed this yet.

4. Key findings and limitations

Only the MCU loop meets a 0.2 ms tolerance under load; SCHED_FIFO meets 1.0 ms but not 0.2 ms DMR.

- **CFS (A) breaks down under load.** DMR at 1.0 ms rises from 0 to 2.76 to 13.91 %, and WCL reaches 6.12 ms. This partly confirms our hypothesis: misses appear, but WCL stayed below the predicted 10 ms.

- **SCHED_FIFO (B) keeps WCL under 0.9 ms** with no misses at 1.0 ms, yet 1.5-1.9 % of cycles miss 0.2 ms at every load level. At idle this fits the idle-state wake-up above; under load the cause is not yet traced (candidate: non-preemptible sections in the stock PREEMPT kernel).

- **The MCU loop (C baseline) stays under 50 µs**, with no misses at 1.0 or 0.2 ms. Linux stress still raised its WCL about 5× (10 → 49 µs), so isolation is strong but not total.

- **Zephyr's clock drifts against the CPU cycle counter.** Our diagnostics show it losing about 19 ppm while busy and 508 ppm while asleep, so the loop really ran at 10.0055 ms per cycle. Jitter and WCL are computed after fitting the period and are unaffected. About half of the 545 ppm offset is still unexplained. The referee board timestamped with the same Zephyr counter and sleeps between pulses, so the clock offset it measured may be mostly its own lost cycles; we are moving it to the DWT counter.

Limitations: one run per condition, so p99.9 rests on about 10 samples. At 0 % stress the Linux threads float across cores; at 50 and 100 % they are pinned to core 0, so load and affinity change together. The kernel is the stock PREEMPT kernel, not PREEMPT_RT. Config C does not yet use IPC.